

Презентация по лабораторной работе №7”

Анализ файловой системы Linux. Команды для работы с файлами и каталогами

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Цель работы

Выполнение лабораторной работы

Вывод

1. Цель работы



Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов.

Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами и работами, по проверке использования диска и обслуживанию файловой системы.

2. Выполнение лабораторной работы



```
mioannis@fedora:~$ touch abc1
mioannis@fedora:~$ cp abc1 april
mioannis@fedora:~$ cp abc1 may
mioannis@fedora:~$ mkdir monthly
mioannis@fedora:~$ cp april may monthly
mioannis@fedora:~$ cp monthly/may monthly/june
mioannis@fedora:~$ ls monthly
april  june  may
mioannis@fedora:~$ mkdir monthly.00
mioannis@fedora:~$ cp -r monthly monthly.00
mioannis@fedora:~$ cp -r monthly.00 /tmp
mioannis@fedora:~$
```

Рисунок 1: Команды из описания лабораторной работы

```
mioannis@fedora:~$ mv april july
mioannis@fedora:~$ mv july monthly.00
mioannis@fedora:~$ ls monthly.00
july  monthly
mioannis@fedora:~$ mv monthly.00 monthly.01
mioannis@fedora:~$ mkdir reports
mioannis@fedora:~$ mv monthly.01 reports
mioannis@fedora:~$ mv reports/monthly.01 reports/monthly
mioannis@fedora:~$ ls
abc1  git-extended  may  monthly  reports  work  Видео
mioannis@fedora:~$
```

Рисунок 2: Команды из описания лабораторной работы

```
mioannis@fedora:~$ touch may
mioannis@fedora:~$ ls -l may
-rw-r--r--. 1 mioannis mioannis 0 окт  3 00:22 may
mioannis@fedora:~$ chmod u+x may
mioannis@fedora:~$ ls -l may
-rwxr--r--. 1 mioannis mioannis 0 окт  3 00:22 may
mioannis@fedora:~$ chmod u-x may
mioannis@fedora:~$ ls -l may
-rw-r--r--. 1 mioannis mioannis 0 окт  3 00:22 may
mioannis@fedora:~$ chmod g-r,o-r monthly
mioannis@fedora:~$ touch abc1
mioannis@fedora:~$ chmod g+w abc1
mioannis@fedora:~$
```

Рисунок 3: Команды из описания лабораторной работы



```
mioannis@fedora:~$ cp /usr/include/sys/io.h ~
mioannis@fedora:~$ mv io.h equipment
mioannis@fedora:~$ mkdir ski.plases
mioannis@fedora:~$ mv equipment ski.plases/
mioannis@fedora:~$ mv ski.plases/equipment ski.plases/equiplist
mioannis@fedora:~$ touch abc1
mioannis@fedora:~$ cp abc1 ski.plases/equiplist2
mioannis@fedora:~$ cd ski.plases/
mioannis@fedora:~/ski.plases$ mkdir equipment
mioannis@fedora:~/ski.plases$ mv equiplist equipment/
mioannis@fedora:~/ski.plases$ mv equiplist2 equipment/
mioannis@fedora:~/ski.plases$ cd
mioannis@fedora:~$ mkdir newdir
mioannis@fedora:~$ mv newbir ski.plases/
mv: не удалось выполнить stat для 'newbir': Нет такого файла или каталога
mioannis@fedora:~$ mv newdir ski.plases/
mioannis@fedora:~$ mv ski.plases/newdir/ ski.plases/plans
mioannis@fedora:~$
```

Рисунок 4: Работа с каталогами



Определим опции команды **chmod**, необходимые для присвоения файлам нужных прав доступа:

- **Australia** — **drwxr--r--**
- **play** — **drwx--x--x**
- **My_oc** — **-r-xr--r--**
- **feathers** — **-rw-rw-r--**

```

mioannis@fedora:~$ mkdir australia play
mioannis@fedora:~$ touch my_os feathers
mioannis@fedora:~$ chmod 711 play
mioannis@fedora:~$ chmod 744 australia
mioannis@fedora:~$ chmod 544 my_os
mioannis@fedora:~$ chmod 664 feathers
mioannis@fedora:~$ ls -l
итого 0
-rw-rw-r--. 1 mioannis mioannis  0 окт  3 00:32 abc1
drwxr--r--. 1 mioannis mioannis  0 окт  3 00:46 australia
-rw-rw-r--. 1 mioannis mioannis  0 окт  3 00:47 feathers
drwxr-xr-x. 1 mioannis mioannis 86 сен 28 22:12 git-extended
-rw-r--r--. 1 mioannis mioannis  0 окт  3 00:22 may
drwx--x--x. 1 mioannis mioannis 24 окт  3 00:15 monthly
-r-xr--r--. 1 mioannis mioannis  0 окт  3 00:47 my_os
drwx--x--x. 1 mioannis mioannis  0 окт  3 00:46 play
drwxr-xr-x. 1 mioannis mioannis 14 окт  3 00:20 reports
drwxr-xr-x. 1 mioannis mioannis 28 окт  3 00:42 ski.places
drwxr-xr-x. 1 mioannis mioannis 10 сен 26 14:37 work
drwxr-xr-x. 1 mioannis mioannis  0 сен 24 20:06 Видео
drwxr-xr-x. 1 mioannis mioannis  0 сен 24 20:06 Документы
drwxr-xr-x. 1 mioannis mioannis 288 окт  1 17:21 Загрузки
drwxr-xr-x. 1 mioannis mioannis  50 сен 26 00:06 Изображения
drwxr-xr-x. 1 mioannis mioannis  0 сен 24 20:06 Музыка
drwxr-xr-x. 1 mioannis mioannis  0 сен 24 20:06 Общедоступные
drwxr-xr-x. 1 mioannis mioannis  0 сен 24 20:06 'Рабочий стол'
drwxr-xr-x. 1 mioannis mioannis  0 сен 24 20:06 Шаблоны
mioannis@fedora:~$

```

Рисунок 5: Изменение прав доступа и просмотр



```
mioannis@fedora:~$ cat /etc/passwd
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/usr/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/usr/sbin/nologin
polkitd:x:114:114>User for polkitd:/usr/sbin/nologin
systemd-coredump:x:997:997:systemd Core Dumper:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/usr/sbin/nologin
chrony:x:995:995:chrony system user:/var/lib/chrony:/usr/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
pcscd:x:993:993:PC/SC Smart Card Daemon:/usr/sbin/nologin
```

Рисунок 6. Содержимое файла /etc/passwd

```
mioannis@fedora:~$ cp feathers file.old
mioannis@fedora:~$ mv file.old play/
mioannis@fedora:~$ mkdir fun
mioannis@fedora:~$ cp -R play/ fun/
mioannis@fedora:~$ mv fun/ play/games
mioannis@fedora:~$ chmod u-r feathers
mioannis@fedora:~$ cat feathers
cat: feathers: Отказано в доступе
mioannis@fedora:~$ cp feathers feathers2
cp: невозможно открыть 'feathers' для чтения: Отказано в доступе
mioannis@fedora:~$ chmod u+r feathers
mioannis@fedora:~$ chmod u-x play/
mioannis@fedora:~$ cd play/
bash: cd: play/: Отказано в доступе
mioannis@fedora:~$ chmod +x play/
mioannis@fedora:~$ man mount
mioannis@fedora:~$ man mkfs
mioannis@fedora:~$ man kill
mioannis@fedora:~$
```

Рисунок 7: Упражнения раздела 4



```

MOUNT(8)                                     System Administration                                MOUNT(8)

NAME
    mount - mount a filesystem

SYNOPSIS
    mount [-h|-V]

    mount [-l] [-t fstype]

    mount -a [-ffnrsvw] [-t fstype] [-O optlist]

    mount [-fnrsvw] [-o options] device mountpoint

    mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

    mount --bind|--rbind|--move olddir newdir

    mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable] mountpoint

DESCRIPTION
    All files accessible in a Unix system are arranged in one big tree, the file hierarchy, rooted at /. These files can be spread out over several devices. The mount command serves to attach the filesystem found on some device to the big file tree. Conversely, the umount(8) command will detach it again. The filesystem is used to control how data is stored on the device or provided in a virtual way by network or other services.

    The standard form of the mount command is:

        mount -t type device dir

    This tells the kernel to attach the filesystem found on device (which is of type type) at the directory dir. The option -t type is optional. The mount command is usually able to detect a filesystem. The root permissions are necessary to mount a filesystem by default. See section "Non-superuser mounts" below for more details. The previous contents (if any) and owner and mode of dir become invisible, and as long as this filesystem remains mounted, the pathname dir refers to the root of the filesystem on device.

```

Рисунок 8: Инструкция к команде mount



```
mkfs(8)                                     System Administration                               mkfs(8)

NAME
  mkfs - build a Linux filesystem

SYNOPSIS
  mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
  This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

  mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The device argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file that shall contain the filesystem. The size argument is the number of blocks to be used for the filesystem.

  The exit status returned by mkfs is 0 on success and 1 on failure.

  In actuality, mkfs is simply a front-end for the various filesystem builders (mkfs.<fstype>) available under Linux. The filesystem-specific builder is searched for via your PATH environment setting only. Please see the filesystem-specific builder manual pages for further details.

OPTIONS
  -t, --type type
    Specify the type of filesystem to be built. If not specified, the default filesystem type (currently ext2) is used.

  fs-options
    Filesystem-specific options to be passed to the real filesystem builder.

  -V, --verbose
    Produce verbose output, including all filesystem-specific commands that are executed. Specifying this option more than once inhibits execution of any filesystem-specific commands. This is really only useful for testing.

  -h, --help
    Display help text and exit.

  -V, --version
    Display version and exit. (Option -V will display version information only when it is the only parameter, otherwise it will work as --verbose.)

BUGS
  All generic options must precede and not be combined with filesystem-specific options. Some filesystem-specific programs do not automatically detect the device size and require the size parameter to be specified.

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Manual page mkfs(8) line 1 (press h for help or q to quit)
```

Рисунок 9: Инструкция к команде mkfs



```
KILL(1) User Commands KILL(1)

NAME
    kill - terminate a process

SYNOPSIS
    kill [-signal|-s signal|-p] [-q value] [-a] [--timeout milliseconds signal] [--] pid/name...

    kill -l [number|0xsigmask] | -L

    kill -d pid

DESCRIPTION
    The command kill sends the specified signal to the specified processes or process groups.

    If no signal is specified, the TERM signal is sent. The default action for this signal is to terminate the process. This signal should be used in preference to the KILL signal (number 9), since a process may install a handler for the TERM signal in order to perform clean-up steps before terminating in an orderly fashion. If a process does not terminate after a TERM signal has been sent, then the KILL signal may be used; be aware that the latter signal cannot be caught, and so does not give the target process the opportunity to perform any clean-up before terminating.

    Most modern shells have a builtin kill command, with a usage rather similar to that of the command described here. The --all, --pid, and --queue options, and the possibility to specify processes by command name, are local extensions.

    If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS
    The list of processes to be signaled can be a mixture of names and PIDs.

    pid
        Each pid can be expressed in one of the following ways:

        n
            where n is larger than 0. The process with PID n is signaled.

        0
            All processes in the current process group are signaled.
```

Рисунок 10: Инструкция к команде kill



```
FSCK(8)                                     System Administration                                     FSCK(8)

NAME
    fsck - check and repair a Linux filesystem

SYNOPSIS
    fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--] [fs-specific-options]

DESCRIPTION
    fsck is used to check and optionally repair one or more Linux filesystems. filesystem can be a device name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a filesystem label or UUID specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or LABEL=root). Normally, the fsck program will try to handle filesystems on different physical disk drives in parallel to reduce the total amount of time needed to check all of them.

    If no filesystems are specified on the command line, and the -A option is not specified, fsck will default to checking filesystems in /etc/fstab serially. This is equivalent to the -As options.

    The exit status returned by fsck is the sum of the following conditions:

    0      No errors

    1      Filesystem errors corrected

    2      System should be rebooted

    4      Filesystem errors left uncorrected
```

Рисунок 11: Инструкция к команде fsck

3. Вывод



В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.